

UAV Initial Sizing — Design Report

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1 · Mission Parameters

Parameter	Value
Cruise speed	80.0 m/s (288 km/h)
Mission altitude	6000 m
Mission range	500 km
Mission endurance	8.0 hr
Payload role	Strike
Mission objective	High Endurance
UAV class	large
Max. load factor	3.00 g
Engine type	Turboprop

2 · Weight Budget

Component	Mass [kg]
MTOW	3146.2
Empty weight	1758.7
Fuel weight	3177.6
Payload weight	1387.5
Fuel fraction W_f/W_0	1.010
Empty fraction W_e/W_0	0.559

3 · Wing & Aerodynamic Sizing

Parameter	Value
Wing area S	38.19 m ²
Wing span b	19.15 m
Aspect ratio AR	9.60
Wing loading W/S	807.9 N/m ²
Cruise L/D	14.53
Power loading W/P [kg/W]	0.09135 (10.9 W/kg)

4 · Fuselage

Parameter	Value
Fuselage length	13.62 m
Fuselage radius	0.817 m
Cylinder start	10.0 %

Cylinder end	70.0 %
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5 · Longitudinal Stability

Parameter	Value
CG position (from nose)	6.182 m
Neutral point (from nose)	7.089 m
Static margin SM	4284.8 % MAC
Assessment	STABLE — SM = 42.8% MAC (target 5-15%)

6 · Design Point — W/P vs W/S Diagram

Wing Loading vs Weight-to-Power Across Altitudes

